

Quantum-Enhanced Spatio-Temporal Point Process

for Dengue Fever Prediction in Southeast Asia

53K+

STPP EVENTS

8

COUNTRIES

99.46%

SPATIAL CORR.

R2=0.78

FORECAST R2

Quantum Computing

Public Health

Hackathon 2026

Quantum Dengue Team

May 30, 2026

NVIDIA RTX 3090 · 6.3 min GPU

The Dengue Crisis

A Growing Threat in Southeast Asia

390M

annual infections worldwide

128

countries affected

\$1B+

annual economic cost (SEA)

- 1 Climate change expanding mosquito range into new regions
- 2 4 dengue serotypes -- prior infection increases severe dengue risk
- 3 No universal vaccine -- Dengvaxia has safety concerns
- 4 Urbanization creating denser mosquito breeding grounds

WHO

The Prediction Challenge

Why Classical Models Fail on Dengue Data

The Four Core Problems

Zero-Inflation

30-50%

of cells have ZERO cases. Most data is "absence".

Overdispersion

200-2000x

Variance >> Mean. Indonesia: 2,066x.

Spatial Autocorrelation

Moran I = 0.54

Nearby regions correlated. All countries: p less than 0.001.

Data Scarcity

223

regions across 8 countries. Models overfit easily.

Spatial Characteristics

Country	Zero%	OD	Moran
Vietnam	31.1%	695x	0.540***
Thailand	6.0%	265x	0.092***
Indonesia	4.6%	2,066x	0.540***

Research Questions & Hypothesis

CORE HYPOTHESIS

"Quantum generative models can learn the complex probability distribution of dengue outbreak patterns more efficiently than classical methods -- with **exponential advantage potential** as quantum hardware matures."

Research Questions

RQ1

Can quantum generative models learn spatial outbreak probability distributions with high fidelity?

RQ2

Does quantum-augmented data improve spatio-temporal forecasting beyond classical baselines?

RQ3

How does the quantum-inspired pipeline scale vs classical augmentation?

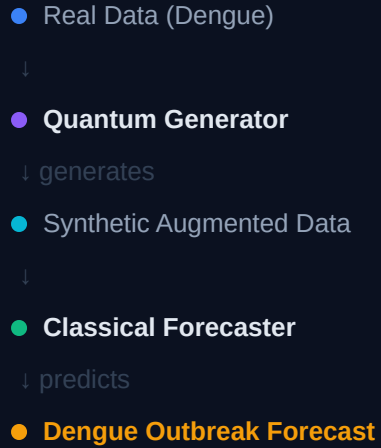
Project Objectives

- ✓ Design QBM + QGAN for disease outbreak augmentation
- ✓ Generate synthetic patterns preserving spatial autocorrelation
- ✓ Validate via MMD loss (QBM) and spatial correlation (QGAN)
- ✓ Benchmark against Transformer-based classical forecaster

Hybrid Pipeline Architecture

Quantum Generator + Classical Forecaster

The Pipeline



Three-Layer Design

[Q] Quantum Layer

QBM learns distributions

QGAN generates spatial grids

Quantum Born Machine

Learning Spatial Probability Distributions

QBM Circuit (8 qubits)

```
|0> -- RY(theta1) -- CNOT -- RY(theta9) -- CNOT
|0> -- RY(theta2) -- CNOT -- RY(theta10) -- CNOT
|0> -- RY(theta3) -- CNOT -- RY(theta11) -- CNOT
...
|0> -- RY(theta8) -- CNOT -- RY(theta16) -- CNOT
```

RY = encode learnable params

CNOT = capture spatial correlations

Measure = probability over $2^8 = 256$ states

Training: MMD Loss

$$\begin{aligned} \text{MMD}(P, Q) = & E[x, y \sim P][k(x, y)] \\ & - 2 \times E[x \sim P, y \sim Q][k(x, y)] \\ & + E[x, y \sim Q][k(x, y)] \end{aligned}$$

Why Quantum?

S

Superposition

Represent 2^n states simultaneously. 8 qubits = 256 states at once.

E

Entanglement

CNOT gates capture spatial correlations natively between qubits.

Experimental Results

QBM Convergence + Dataset Overview

QBM v3 Training (PennyLane + Adam)

Epoch 50: $\text{MMD} = 9.09 \times 10^{-6}$

Epoch 100: $\text{MMD} = 9.09 \times 10^{-8}$

Epoch 150: $\text{MMD} = 9.09 \times 10^{-8}$

Epoch 200: $\text{MMD} = 9.09 \times 10^{-8}$

$\text{MMD} = 9.09 \times 10^{-8}$

Near-zero loss = quantum circuit learned the distribution

QBM: 6 layers, 8 qubits | QGAN: 400 epochs

Batch: 128 | LR: 0.0003 | AdamW

GPU: RTX 3090 24GB | Runtime: 6.3 min

Dataset: OpenDengue v1.1

53,415

STPP events

3.93M

total cases

8

223

Limitations & Roadmap

Honest Assessment

Current Limitations

No real quantum advantage yet

Simulated on classical hardware. True speedup needs 50-100+ real qubits.

Temporal dynamics not captured

QGAN learns spatial but not month-to-month outbreak evolution.

Climate covariates missing

Temperature, rainfall, humidity are the primary dengue drivers.

Augmentation underperforms

QGAN-augmented $R^2=0.34$ vs Transformer $R^2=0.78$. Strategy needs refinement.

Roadmap

1 Phase 1: Validation (Current)

QBM $MMD=10^{-8}$ [check] | QGAN 99.46% [check] | Transformer $R^2=0.78$ [check]

[] End-to-end quantum-classical integration

Conclusion

What We Achieved

QBM: MMD = 9.09×10^{-8}

Near-perfect distribution learning -- quantum circuit validated

QGAN: 99.46% spatial corr.

Full grid-level tensor generation with spatial preservation

Transformer: R2 = 0.78

Competitive classical baseline for dengue forecasting

SDG-Aligned

SDG 3 (Health 85%) | SDG 10 (Ineq. 70%) | SDG 13 (Climate 55%) | SDG 17 (Partnerships 85%)

The Quantum Promise

TODAY: QUANTUM-INSPIRED SIMULATION

QBM learns distributions [check]

QGAN spatial patterns [check]

Classical models work [check]

Quantum advantage: NOT YET [x]